CESNET S3 & Kubernetes Manuals

Pollination Ecology Group (Handrkov) — Practical Handbook & Templates

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Welcome

Welcome to the practical manuals for using **CESNET Object Storage (S3)** and **CESNET / e-INFRA CZ Kubernetes (CERIT-SC)** in our research group.

These guides are written for all members of the lab—whether you want to browse or upload field datasets using a graphical interface on Windows/macOS, read `.rds`/`.csv` files directly into R or Python, run heavy CPU/GPU scripts on the cluster without building Docker containers, or deploy interactive web applications.

Manual Suite Overview

Manual	Markdown	HTML	PDF	What It Covers
1. CESNET S3 Storage	01_CESNET_S3_Storage_Manual.md	HTML	PDF	GUI access (Cyberduck / WinSCP), mounting as a local folder (`s3fs`), fast bulk transfers (`rclone` / `aws-cli`), and direct **R** (`aws.s3`) & **Python** (`boto3`) integration.
2. Kubernetes Getting Started	02_Kubernetes_Getting_Started.md	HTML	PDF	Rancher login (`rancher.cloud.e-infra.cz`), `kubeconfig.yaml` setup, installing `kubectl`, mandatory CERIT-SC non-root security rules, and the `kubectl` cheat sheet.
3. Running R & Python Scripts	03_Running_R_and_Python_Scripts_on_K8s.md	HTML	PDF	Running `.R` and `.py` scripts on CPU or NVIDIA GPUs **without building Docker images** (`ConfigMap` + `Secret`), S3 data syncing, and interactive debugging pods.
4. Deploying Web Apps	04_Deploying_Web_Apps_on_K8s.md	HTML	PDF	Hosting Flask/Waitress, Streamlit, or R Shiny dashboards with automatic HTTPS (`*.dyn.cloud.e-infra.cz`) and zero-downtime updates.

Ready-to-Use Templates (`templates/`)

You do not need to write configuration files from scratch. Copy and adapt the templates in the `templates/` folder:

S3 Storage Templates (`templates/s3/`)

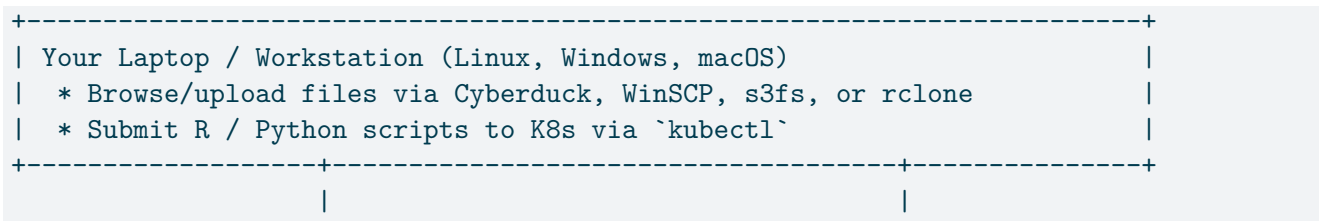
- **`templates/s3/mount_cesnet_s3.sh`** — Mounts or unmounts the `peg-bucket` S3 storage as a local folder (`~/cesnet_s3`) on Linux/macOS via `s3fs`.
- **`templates/s3/rclone_backup_template.sh`** — High-speed parallel upload/backup script using `rclone copy` (safe against accidental deletions).
- **`templates/s3/s3_example.R`** — Complete R script showing how to list files, read/save `.rds` and `.csv` objects directly in memory, and upload/download files using `aws.s3`.
- **`templates/s3/s3_example.py`** — Complete Python script using `boto3` and `pandas` to list, stream, download, and upload files to `peg-bucket`.

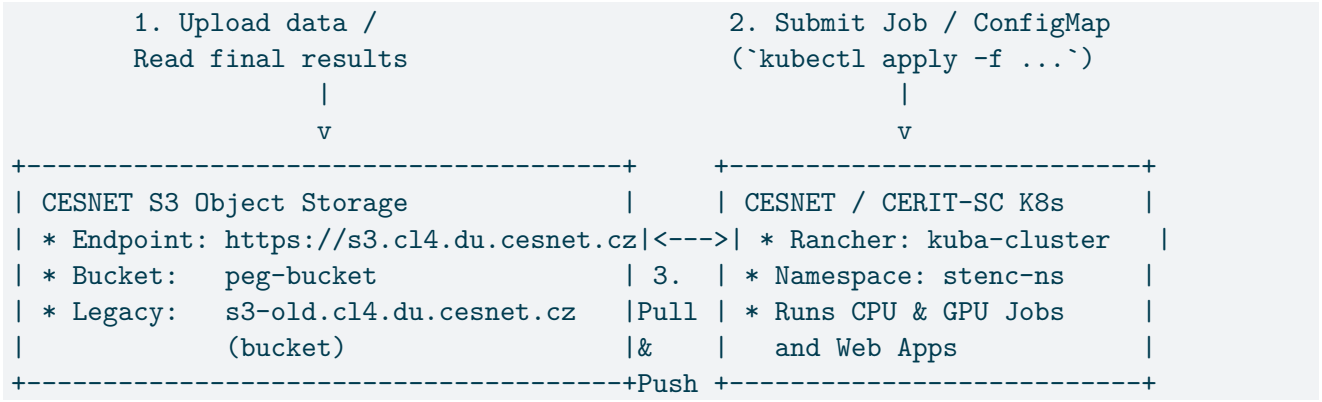
Kubernetes Templates (`templates/k8s/`)

- **`templates/k8s/run_script_on_k8s.sh`** — **One-command launcher**: pass any local `.R` or `.py` file (`./run_script_on_k8s.sh my_script.R` or `./run_script_on_k8s.sh train.py --gpu`) and it automatically uploads the script to a `ConfigMap`, submits a CERIT-SC compliant Job with S3 credentials, and streams live logs!
- **`templates/k8s/r-cpu-job.yaml`** — YAML template for running R scripts (`rocker/tidyverse:4.4.1`) with S3 credentials and writable user library directory.
- **`templates/k8s/python-cpu-job.yaml`** — YAML template for running Python CPU data-processing jobs (`python:3.12-slim`).
- **`templates/k8s/python-gpu-job.yaml`** — YAML template for running GPU deep learning jobs (`ultralytics/ultralytics:latest` / PyTorch) with 4 GiB `/dev/shm` shared memory and S3 sync.
- **`templates/k8s/web-app-deployment.yaml`** — Complete `Deployment` + `Service` + HTTPS `Ingress` manifest for hosting web applications on `*.dyn.cloud.e-infra.cz`.

Architecture Overview: How CESNET S3 & Kubernetes Work Together

In our workflow, **Kubernetes compute pods are stateless** (when a job finishes, its temporary container disk disappears). Therefore, all persistent datasets, images, and model outputs live on **CESNET S3 Object Storage**:





Quick Reference Table (Group Defaults)

Parameter	Value	Notes
Primary S3 Endpoint	`https://s3.cl4.du.cesnet.cz`	Current production CESNET CL4 Ceph S3 endpoint
Primary S3 Bucket	`peg-bucket`	Pollination Ecology Group shared bucket
Legacy S3 Endpoint / Bucket	`https://s3-old.cl4.du.cesnet.cz` / `bucket`	Older storage cluster (use only if migrating legacy files)
Rancher Web Dashboard	`https://rancher.cloud.e-infra.cz`	Log in via eduID.cz (Charles University / CAS)
Kubernetes Cluster	`kuba-cluster`	Download `kubeconfig.yaml` from Rancher
Kubernetes Namespace	`stenc-ns`	Default namespace for our group's jobs, secrets, and apps
K8s S3 Credentials Secret	`s3-credentials`	Pre-configured Secret in `stenc-ns` (`access-key` & `secret-key`)